

■ SLO-1 Department-Level Intervention: Problem Analysis

SLO-1: *An ability to identify, formulate, and solve complex engineering problems by applying principles of engineering, science, and mathematics.*

All courses contribute evidence by mapping one assignment/exam question to this rubric. Focus: Checking whether students can decompose problems with the Given–Find–Assumptions–Approach–Units framework.

Overlay Rubric: Quick Steps

- Grade normally (no change to your grading).
 - **Pick ONE assignment item** that aligns to the SLO/PI (**not the whole exam average**).
- Example (SLO-1 / PC2): Use Midterm Problem #3 where students set up equations with assumptions.
- If Problem #3 is graded out of 20 points and a student earns 15/20, enter 75 in the Score column of the accompanying Excel sheet.
 - ENTER 75 (not 75%). Use whole numbers 0–100 only.
 - The 'Rubric Level' column classifies scores into Novice/Apprentice/Practitioner/Expert.
 - The Summary sheet shows the PERCENTAGE of students at each level and a TOTAL that should equal 100%.

Performance Indicator (PI)	Level	Lower-Level Expectation (Sophomore)	Upper-Level Expectation (Senior/Capstone)	Bloom's Link
PC1: Identify Engineering Problem	Novice	Misinterprets or overlooks key aspects of a structured problem.	Misinterprets or oversimplifies an open-ended problem.	Understand
	Apprentice	Recognizes general topic but misses scope or details.	Identifies main issue but ignores important context/constraints.	Apply
	Practitioner	Correctly identifies the problem in a structured, well-defined scenario.	Accurately identifies and frames an ill-structured, open-ended problem, including scope and context.	Apply → Analyze
	Expert	Goes beyond restating; identifies relevance in course context.	Frames problem comprehensively; connects to real engineering/scientific context; highlights significance.	Analyze
PC2: Formulate a Problem	Novice	Minimal or no attempt to structure; assumptions missing.	Minimal attempt; overlooks ambiguity and constraints.	Apply
	Apprentice	Attempts formulation but incomplete; checklist partially used.	Formulates partially; assumptions weakly justified under complex conditions.	Apply
	Practitioner	Uses Given–Find–Assumptions–Approach–Units checklist with most elements correct.	Fully applies checklist, reasonably justifies assumptions, decomposes problem under moderate ambiguity.	Analyze
	Expert	Uses checklist completely; assumptions explicit.	Comprehensive formulation with all checklist elements; assumptions critically justified and connected across domains.	Analyze → Evaluate
PC3: Solve a Problem	Novice	Incorrect or incomplete solution for structured problems.	Incorrect or incomplete solution for multi-step/open-ended problems.	Apply
	Apprentice	Partial solution; applies some relevant principles with gaps.	Partial solution; applies relevant concepts but misses integration across domains.	Apply
	Practitioner	Mostly correct solution for single-step structured problems; logical process.	Mostly correct solution for multi-step problems; integrates checks (units, reasonableness).	Analyze
	Expert	Correct and justified solution; clear reasoning.	Elegant, well-justified solution with synthesis across multiple topics; includes sensitivity/validation checks.	Analyze → Evaluate